

Planning-Guided Failure-Triggered Recovery via Imitation Learning for Dynamic Social Navigation

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Abstract—Classical navigation stacks are reliable in routine mapped environments but can fail in reciprocal interactions with moving people. End-to-end learned navigation can address such interactions, yet replacing a mature planner everywhere changes behavior even where no recovery is needed. We study a bounded alternative: an observable failure detector invokes a learned policy only during collision risk, freezing, oscillation, or deadlock. The policy chooses among 21 temporary subgoals and four interpretable modes, while a planning-derived mask rejects invalid actions and Nav2 executes each selected goal. A privileged short-horizon expert supplies imitation labels and DAGger covers learner-induced recovery states. During validation we discovered that accepted pose updates did not move collision geometry declared static. We converted fallback pedestrians to kinematic links, committed privileged state only after confirmed geometry updates, and added a privileged-to-LiDAR validity gate. Across one fixed validation seed at each of three crossing-flow densities, DWB collided in all three episodes, whereas recovery reached the original goal in all three. This six-episode precheck establishes corrected closed-loop execution, not statistical superiority; multi-seed evaluation remains required.

I. Introduction

Classical navigation offers explicit maps, reusable planners, and predictable nominal behavior, but local optimization can freeze or oscillate in interactive bottlenecks. Learned policies can encode interaction patterns, but continuous end-to-end replacement expands the learned component’s authority and makes failures harder to interpret.

We instead confine learning to the failure boundary. Figure 1 shows the hierarchy. A classical planner owns nominal PointGoal navigation. An observable detector and hysteretic state machine trigger a bounded recovery option. A compact policy chooses an interpretable temporary target or mode from a fixed action set, subject to one mask shared by data generation and deployment. Once progress resumes, the original task goal is reissued.

The current contribution is threefold: (i) a failure-triggered architecture that preserves classical nominal control; (ii) an automated privileged planning expert that labels masked temporary-goal choices; and (iii) an imitation pipeline using behavior cloning and DAGger. We do not claim a first recovery system or a formal safety guarantee. This evidence-linked draft reports the corrected simulator-validity result and does not reuse pre-kinematic comparisons as performance evidence.

II. Related Work

Arena-Bench and Arena-Rosnav provide reproducible interfaces for comparing model-based and learned navigation in dynamic environments [1], [2], [3], [4]. All-in-One learns to select among navigation planners [5]; DR-MPC learns a residual around model-predictive control [6]. Both motivate hybrid navigation but differ from failure-only intervention.

Learned navigation recovery predates this work. Del Duchetto et al. learn failure detection and local recovery from human demonstrations [7]. Mohammad et al. proactively predict planning failure and search for a recovery state [8]. Recent failure-aware learning uses unsafe experience to shape value estimation while restricting policy learning to successful demonstrations [9]. Our narrower distinction is automatic recovery demonstrations from privileged rollout planning, temporary-subgoal decisions, and aggregation on learner-induced recovery states. DAGger is the standard mechanism for mitigating sequential distribution shift [10].

III. Method

A. Failure-Triggered Hierarchy

Let g be the original goal and let the classical stack produce command u_b . From observable history o_t , the detector returns scores for collision risk, freeze, oscillation, and deadlock. Recovery begins after the maximum score exceeds τ_{on} and exits only after it remains below $\tau_{\text{off}} < \tau_{\text{on}}$ while valid goal progress resumes. Cooldown, option duration, and recovery-count limits prevent rapid switching.

B. Action Space and Mask

The action space contains temporary goals at radii $\{0.6, 1.0, 1.4\}$ m and bearings $\{-90, -60, -30, 0, 30, 60, 90\}^\circ$, plus WAIT, BACKUP, REPLAN, and CONTINUE. The mask rejects occupied, disconnected, LiDAR-occluded, rear-unobserved, or path-divergent motion. During latched collision risk it also blocks unsafe continuation and applies endpoint and swept-segment clearance. Masking precedes softmax, so invalid actions cannot be selected while a valid fallback remains.

An independent stopping layer uses

$$d_{\text{stop}} = \frac{v^2}{2a_{\text{brake}}} + v\ell + m, \quad (1)$$

original goal in 108.9, 117.9, and 125.4 s, respectively, with minimum human-centre distances of 1.166, 1.151, and 0.936 m. It used 150, 163, and 439 recovery samples, revealing increasing intervention cost with density. These pairs verify execution and motivate a frozen multi-seed pilot; they are too few for a significance claim.

VI. Limitations

The method assumes a known two-dimensional map and planar LiDAR. Its privileged expert depends on simulator state and constant-velocity pedestrian prediction. The finite discrete action set cannot express arbitrary maneuvers. The Gazebo kinematic fallback differs from real crowds and exhibits scheduling artifacts. There is no formal safety guarantee, hardware study, completed learned detector, or PPO refinement. Corrected training data and the final multi-scenario test remain incomplete.

VII. Conclusion

Failure-triggered temporary-goal recovery can be integrated with a classical ROS2 navigation stack and trained from an automated privileged planner without granting the learned policy continuous velocity authority. The simulator-validity gate exposed and corrected a hidden proxy error. Under the corrected implementation, the learned hierarchy converted three crossing-flow DWB collisions into three goal reaches across density levels. This small precheck is promising but not inferential evidence. The required next steps are corrected-data retraining and a locked paired multi-scenario evaluation.

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